AN EMBEDDED PROGRAM FOR AUTOMATIC HEADLIGHT INTENSITY ADJUSTMENT

(SMARTBEAM)

BY

Kabagyo Desire Acquilina
23/U/1626/GCS/PS

Lanyero Fortunate
23/U/3718/GCS/PS

Akejo Ivan Gira
23/U/1605/GCS/PS

A Report Submitted to the Department of Computer Science, Faculty of Science for the Study in Partial Fulfillment of the Award of Bachelor's Degree of Science in Computer Science at Gulu University

June, 2026

Declaration

We, SmartBeam group, hereby declare that whatever is presented in this report is our own work and it has never been submitted to any academic institution of higher learning and by any group in this same or previous academic years. This report contents have been compiled with generous academic integrity following the parameters and guidelines presented and organized by the university and any academic decision taken based on the report contents is however based on the individual responsibility.

We also affirm that the information presented in this report, including the project findings, conclusions and recommendations, accurately reflects the work conducted by our group during the project.

KABAGYO DESIRE ACQUILINA ....

LANYERO FORTUNATE ....

AKEJO IVAN GIRA ....

Approval

This project report has been submitted for examination with approval of the supervisor Assoc.Prof. Benedict Oyo.

Sign: .... Date: ....

Department of Computer Science
Gulu University

Dedication

We, the members of Group SmartBeam, dedicate this report to Gulu University, our esteemed institution, whose nurturing environment has fostered our learning, growth, and intellectual exploration.

This work is also dedicated to our families and loved ones. Their unconditional love, encouragement, and patience have anchored us through every challenge, and their belief in our abilities has continually motivated us to strive for excellence.

Finally, we dedicate this report to the drivers and motorists of Uganda, whose experiences with night-time driving hazards and headlight glare inspired this project. It is our sincere hope that SmartBeam will, in its own humble way, make night travel safer, reduce glare-related accidents, and contribute to the well-being of commuters in Gulu and across the nation.

Acknowledgments

We extend our profound gratitude to all individuals and organizations whose contributions made this report possible. First, we thank the research team for their unwavering dedication and rigorous problem-solving throughout this demanding project. We are especially grateful to our project supervisor, Assoc. Prof. Benedict Oyo and , whose insightful guidance, technical expertise, and encouragement shaped the direction and quality of this research, and our mentor, Mr Allan Junior Asiku whose unwavering guidance and support has shaped our project development journey and inspired our technological curiosity. We also sincerely thank the faculty members of the Department of Computer Science at Gulu University for fostering a nurturing academic environment that supported our intellectual growth especially Dr. Christine Kalumera. Our appreciation extends to Gulu University itself for providing the essential resources and facilities that enabled our work.

Special thanks go to the local drivers and mechanics who participated in our interviews and data collection. Their real-world experiences with night-time driving hazards and invaluable technical feedback were absolutely vital to developing, calibrating, and validating the SmartBeam prototype.

Finally, we express heartfelt appreciation to our families, whose enduring love and encouragement formed the foundation of our academic journey. To all who contributed, directly or indirectly, we offer our sincere thanks, your support was indispensable to the successful realization of this project.

Contents

List of Tables

List of Figures

List of Acronyms

Advanced RISC Machine -M series	ARM Cortex
Controller Area Network	CAN
Electric Control Unit	ECU
Insurance institutes of highway safety	IIHS
Light emitting diodes	LED
Microcontroller unit	MCU
Sport unity vehicle	SUV
Analog-to-Digital Converter	ADC
Light Dependent resistor	LDR

Abstract

Night-time driving poses significant safety risks due to headlight glare from oncoming vehicles, which often causes temporary blindness, reduced visibility, and increased accident rates. While modern vehicles incorporate advanced adaptive lighting systems, these technologies remain expensive and inaccessible to the majority of vehicles on Ugandan roads.

This project presents the design and development of SmartBeam, an affordable embedded system for automatic headlight intensity adjustment. The system utilizes an Arduino uno, light sensors to detect oncoming headlights and taillights in real-time and dynamically adjusts the host vehicle's headlight intensity (high beam, low beam, and intermediate levels) to minimize glare while maintaining optimal road illumination.

The project followed the Systems Development Life Cycle (SDLC) methodology, encompassing requirements analysis, system design, hardware prototyping, software implementation, and testing. The prototype was developed using affordable components including an Arduino uno board, relay module for beam adjustment and a white light lamp to simulate vehicle headlights.

The SmartBeam system aims to provide a cost-effective retrofit solution suitable for older passenger vehicles such as sedans and SUVs commonly used in Uganda. By reducing glare-related accidents and improving driver comfort and visibility during night-time travel, this project contributes to enhanced road safety in resource-constrained environments. The successful implementation of this embedded program demonstrates the potential of accessible intelligent transportation systems in addressing local road safety challenges

Chapter 1

Introduction

1.1 Background

Night-time driving is preferred by many drivers because fewer people are on the road, resulting in less traffic and more favourable temperatures for long-distance journeys. However, it is inherently risky due to irresponsible behaviours, such as drunk driving. Additionally, visibility issues arise because the human eye struggles to adjust rapidly from bright light to darkness, which can lead to temporary blindness when drivers look directly into oncoming headlights. Furthermore, because night-time travel often occurs at the end of the day, driver fatigue and drowsiness are inevitable and remain major causes of night-time accidents. Excessive illumination stemming from a combination of vehicle headlights and road lights can also impair vision, making it difficult for drivers to see clearly. Consequently, the issue of night-time visibility and the associated risks of accidents caused by overly bright headlights have become significant concerns in road safety.

Approximately half of all traffic deaths occur between 8:00 p.m. and 11:59 p.m., highlighting the critical importance of effective headlights (Joseph et al., 2026). Evaluations by the Insurance Institute for Highway Safety (IIHS) indicate that on-road illumination provided by vehicle lights varies significantly; only a small percentage of headlight systems receive a "good" rating for visibility and glare control (Raychaudhuri et al., 2025). This performance variation emphasizes the need for more advanced solutions to enhance nighttime driving safety. Innovative technologies, such as high-beam assist, curve-adaptive headlights, and adaptive driving beams, have been developed to improve visibility and reduce glare (Zhou et al., 2025). However, despite these advancements, a significant gap remains in the effectiveness of these technologies in addressing the challenges of glare and visibility during night-time driving. This paper proposes an innovative solution: the development of an embedded program for automatic headlight intensity adjustment. By utilizing advanced sensor technologies, this program aims to dynamically adjust headlight intensity in real-time, thereby reducing glare and improving overall road safety during night-time travel. This approach addresses the immediate challenge of glare while contributing to the broader goal of enhancing road safety through technological application. The rapid development of intelligent automobile technology including real-time adjustment systems for headlamp intensity based on sensor feedback has opened new avenues for improving the safety and driving experience at night (Naskar & Saha, 2025; Zhou et al., 2025).

These systems leverage data from various sensors, such as cameras, radars, and light sensors, to evaluate lighting requirements in real-time, adjusting brightness to provide optimal illumination for current driving conditions (Li et al., 2025). Ultimately, this research demonstrates how technology can significantly improve night-time driving safety by mitigating excessive brightness and enhancing visibility.

1.2 Statement of the Problem

Night-time driving poses significant challenges to road safety, including temporary blindness caused by excessive headlight glare, visibility issues stemming from the human eye's slow adjustment between bright light and darkness, and heightened accident risks due to driver fatigue (Abdulrahman et al., 2025). Modern vehicles, such as the Toyota Harrier (2018+), Subaru Impreza (2017+), and Nissan X-Trail (2018+), feature adaptive high-beam assist systems that automatically modulate light intensity based on the environment; for instance, detecting oncoming traffic and dimming lights to preserve visibility for both drivers. However, such advanced vehicles likely constitute less than 5% of the fleet in Uganda, necessitating an affordable, automated headlight adjustment solution to enhance safety for the broader driving population.

1.3 Main Objective

The purpose of this research is to develop an embedded program for automatic headlight intensity adjustment to enhance night time driving safety.

1.4 Specific Objectives

(i) To investigate the current effectiveness of existing technologies, such as high-beam assist, curve-adaptive headlights, and adaptive driving beams

(ii) To design an embedded program that uses advanced sensor technology to dynamically adjust the headlight intensity based on real-time.

(iii) To implement the designed embedded program for automatic headlight adjustment.

(iv) To test the effectiveness of the embedded program in reducing glare and improving visibility during night time driving.

1.5 Scope

This project focuses on developing an embedded system for automatic headlight intensity adjustment, targeting passenger vehicles commonly used in Uganda, such as sedans and SUVs. The functionality includes real-time sensor-based detection of oncoming headlights and taillights, dynamic intensity control (high/low beam with intermediate levels). Geographically, the study is centred on Uganda, with field trials conducted in Gulu University. Methodologically, this project employs the Systems Development Life Cycle (SDLC) up to the testing phase, using affordable hardware like Arduino Uno boards and open-source software tools (Arduino IDE).

1.6 Significance of the study

This project holds substantial academic and practical value by advancing the integration of embedded systems in automotive safety, contributing new knowledge to the field of computer science and intelligent transportation systems. Academically, it explores realtime sensor fusion and algorithm development, potentially inspiring further research in adaptive technologies for resource-constrained environments. Practically, the solution addresses a critical road safety gap in Uganda, where night-time accidents contribute significantly to mortality rates, offering an affordable retrofit for older vehicles to reduce glare-related crashes.

Socio-economically, it could lower healthcare costs associated with accidents, improve driver confidence, and promote safer night-time travel, benefiting communities reliant on roads.

Chapter 2

Literature Review

2.1 Introduction

Headlights of vehicles pose a great danger during night driving. The drivers of most vehicles use high, bright beams while driving at night. This causes discomfort to the driver travelling from the opposite direction who experiences a sudden glare caused by the high intense headlight beam from the vehicle coming towards them. The recent survey reveals that the accidents percentage is high between 8pm to midnight compared to the day time. The top most reason for these accidents is temporary blindness of the drivers caused by the headlights of the opposite side vehicles. Another study shows that the accident chances are high during raining or foggy season than a normal day. To avoid these accidents, we are mainly depending on the fog lights. There are few technologies already existing with the luxury cars but these are highly expensive to be adopted in the economy cars.

This proposal's main objective is to design an economic Automatic Headlight Adjustment system which can be used by the automobile industry.

2.2 State-of-the-art

One of the reviewed approaches has an orientation control system for the headlights which actually rotates the right and the left low beams independently and makes the beam parallel to the curved roads as much as possible for the better visibility in the curved, narrow or mountain roads. In these kinds of systems, they have used two systems, one is for the vehicle and road and the second system is the actual head light system. The second system takes the input from the first system and based on the details it does few calculations and prepares commands based on these calculations and sends to the motor connected with the head light (Akbar Gozali et al., 2025)

There are few systems proposed by the researchers which adjust automatically the direction of optical axes of the headlight using the spherical sensors comprised of metal balls surrounded by the fluid encapsulated with the spherical sensors and also connected to the spherical sensor system. This system is also using the computer-controlled unit, which is kept behind the headlight very closely. Hence the metal ball cooperates with the sensor which is part of the spherical sensor system. This will make the head light function along with the car movement direction (Moradloo et al., 2025)

In another research, researchers proposed an Electric Control Unit (ECU) system for the headlights of the cars or trucks which gets the outputs of the sensors arrangement that is supplied to the above specified electric control unit using a bus for the communication purpose. In this research, the system calculates the value of a radius of the vehicles when it turns on the road which is used further for changing the direction of the optical axes of the swivel lights. The values are computed based on the output of the sensors which are mentioned above. These values go to the ECU and the calculations give the value to adjust the optical axis. (Xie et al., 2025). The ECU actually calculates the optical axis angle to be controlled using the normal values calculated in ECU. These values are used to activate the actuators which plays a major role in changing the direction of the optical axis of the headlights.

2.3 Related Systems

Automatic headlight systems have evolved over the years, offering different levels of functionality to cater to various driving needs. Below are the main examples of automatic headlight systems.

2.3.1 Basic Automatic Headlights

Basic automatic headlights represent the foundational baseline for automated vehicle lighting. The core objective of this system is to manage visibility transitions during twilight or when entering tunnels, eliminating the risk of drivers forgetting to activate their running lights manually (Balaji, 2020).

The system relies on a dashboard-mounted twilight sensor typically a photoresistor or photodiode configured to measure global ambient lux levels. When ambient light drops below a pre-programmed threshold (such as at sunset), the sensor's change in electrical resistance signals the body control module (BCM) to close an electromechanical relay, turning on the low-beam headlamps and taillights.

System Limitations: This architecture is strictly binary and purely reactive. It cannot assess oncoming glare, change beam direction, or adjust intensity. It is completely blind to traffic patterns and only reads static ambient light levels.

2.3.2 Adaptive Headlights

Adaptive headlights move beyond static lighting by active positioning, altering the physical orientation and footprint of the light beam in real time based on the vehicle's dynamic state to optimize safety on curved infrastructure (Musthafa et al., 2017).

Working Mechanism: This system integrates heavily with the vehicle's internal controller area network (CAN bus). It continuously samples inputs from the steering wheel angle sensor, wheel speed sensors, and yaw-rate gyro-sensors. Using these metrics, an onboard electronic control unit (ECU) calculates the driver's intended path and commands tiny, internal stepper motor actuators to swivel the projector lens housings up to 15 degrees into a curve.

Because this system relies on physical moving parts (servo motors, gears, and swiveling brackets), it is susceptible to mechanical wear over time. Additionally, its adjustments are based entirely on host vehicle dynamics rather than the positions of surrounding motorists.

2.3.3 Auto High Beam Assist

Auto High Beam Assist systems automate beam switching on dark highways, maximizing the distance a driver can see while actively preventing temporary blindness for oncoming motorists (Balaji, 2020).

Unlike basic systems that use simple light sensors, High Beam Assist uses a forward-facing CMOS optical camera mounted behind the rearview mirror. This camera runs specialized machine vision algorithms to detect specific light signatures: red wavelengths indicate the taillights of a vehicle ahead, while bright white spots pinpoint oncoming headlamps. When these signatures are detected, the controller triggers a relay to immediately drop the high beam down to a low beam.

The system operates on a simple, binary all-or-nothing approach. When oncoming traffic is detected, the entire high beam cuts out, temporarily reducing the host driver's long-range visibility on the peripheral sides of the road.

2.3.4 Matrix Headlights

Matrix headlights represent the current pinnacle of smart automotive illumination. They completely eliminate the traditional compromise between maximum road visibility and oncoming glare by using highly targeted, selective dimming (Musthafa et al., 2017).

Instead of a single bulb or projector unit, a matrix headlamp features a grid configuration of up to dozens of individually addressable, high-power LEDs. A high-resolution forward camera tracks the exact coordinates of every vehicle on the road. The system's computer processes these coordinates in real time and turns off or dims only the specific LEDs pointing directly at those vehicles. This creates a moving "shadow box" or dark mask around oncoming traffic while keeping the rest of the road brightly lit with full high-beam power. - System Limitations: Matrix systems demand massive computing power for real-time video processing and thermal management for the dense LED clusters. Due to their high hardware complexity and production costs, they are currently limited to premium vehicle tiers and high-end factory installations.

2.3.5 Strengths and weaknesses of the current systems

Table 2.1: Strengths and weaknesses of the current systems

System name	**Strengths**	**Weakness**
Basic Automatic Headlights	- Convenience: Eliminates the need for manual switching. - Safety: Ensures headlights are used when needed, reducing accidents. - Battery Efficiency: Prevents leaving lights on when not required (Balaji, 2020).	- Lack of customization: Limited ability to adjust sensitivity or behaviour. - No advanced features like beam adjustment for traffic or curves (Balaji, 2020).
Adaptive Headlights	- Improved visibility on winding roads and sharp turns. - Reduced glare for oncoming drivers. - Enhanced safety during night-time or poor weather driving (Musthafa et al., 2017).	- Higher cost compared to basic systems. - Complex mechanics that may require more maintenance (Musthafa et al., 2017).
Auto High Beam Assist	- Improves night-time visibility. - Enhances safety for all road users. - Reduces the need for constant manual adjustments (Balaji, 2020).	- It may not always detect smaller vehicles or bicycles. - Reduced effectiveness in heavy rain or fog (Balaji, 2020).
Matrix Headlights	- Superior visibility and precision. - Enhanced safety in complex driving environments. - Customizable beam patterns for different scenarios (Musthafa et al., 2017).	- Expensive to install and repair. - Currently limited to highend vehicle segments (Musthafa et al., 2017).
SmartBeam Headlight	- Eliminates the need for manual switching. - Ensures headlights are used when needed, reducing accidents. - Prevents leaving lights on when not required.	- No advanced features like beam adjustment for traffic or curves.

Chapter 3

Methodology

3.1 Introduction

This chapter explains step-by-step the way embedded program for automatic headlight intensity adjustment was developed. Several established software development methodologies provide different frameworks for managing a project like agile, software development life cycle, waterfall, prototyping and others.

3.2 Potential methodologies

3.2.1 Agile methodology

Represent a flexible, people focused way of developing software and managing projects that emphasizes adaptability, frequent delivery of working increments, close collaboration, and continuous improvement (Alamri et al., 2024). The project's fixed academic deadline and predefined scope conflict with Agile's open ended and change embracing approach, potentially causing scope creep and risking failure to meet the submission deadline

3.2.2 Waterfall

This is a traditional, linear, and sequential approach to project management, especially in software development. It divides the entire process into distinct phases that flow downward like a waterfall such as requirements gathering, system design, implementation (coding), testing (verification), deployment, and maintenance where each phase must be fully completed and approved before the next one begins, with no overlapping or easy backtracking.

3.2.3 System Development Life Cycle (SDLC)

The SDLC is a structured framework that defines the stages involved in developing an information system, from initial feasibility study through maintenance. It provides a systematic, phased approach to software and system development, typically including planning, analysis, design, implementation, testing, deployment, and maintenance. While SDLC offers a disciplined and well-documented process, its traditional linear interpretation can be inflexible and slow to adapt to emergent technical challenges. For embedded systems with tight hardware-software integration, a purely phase-gated SDLC may delay the detection of integration issues, as hardware and software are often developed in parallel and require early validation. Additionally, the formal documentation requirements of a full SDLC can be overhead-heavy for a time-constrained academic project.

3.2.4 Prototyping

This is an iterative approach to software or system development where an early, simplified working version (prototype) of the product is quickly built and presented to users or stakeholders for feedback. Instead of fully defining all requirements upfront, the process starts with basic requirements, creates a preliminary prototype, tests it, gathers input on usability, functionality, and improvements, and then refines or rebuilds it repeatedly until it meets expectations often leading to the final system. This method is especially useful when requirements are unclear or evolving, as it reduces risks, improves user satisfaction through early involvement, and allows for faster detection of issues before heavy investment in full development

3.3 Adopted methodology

To develop the Integrated Control System for Automatic Headlight Intensity Adjustment (Smart-Beam prototype) efficiently and adaptively, the project adopted the Prototyping Model within the broader Software/Hardware Development Life Cycle. This iterative approach is chosen because

(a) Requirements for automatic headlight detection and intensity adjustment were initially uncertain e.g., sensor thresholds, false positives from ambient lights/rain, optimal response latency.

(b) Early working versions allowed rapid validation of sensor fusion, algorithm logic, and hardware integration.

(c) It supported frequent supervisor feedback, backtracking on design choices, and progressive refinement toward a testable prototype.

(d) The fixed academic timeline and resource constraints made rigid linear models e.g., Waterfall. Unsuitable, while full Agile ceremonies add unnecessary overhead.

The development followed an iterative prototyping cycle, repeated 3 times depending on feedback and risk reduction, progressing from basic proof of concept to a functional prototype suitable for lab and limited field trials (Tucker, 2025).

3.3.1 Phase 1: Initial Requirements Identification & Planning

The data collection process commenced after the researchers obtained an official introduction letter from the Faculty of Science, Gulu University. This letter was presented to the selected participants to explain the purpose of the study and seek their cooperation. Following the acquisition of permission, the researchers identified potential participants, comprising drivers and mechanics who were relevant to the study, and scheduled face-to-face interview sessions at times and locations convenient for them. During these interviews, participants were informed about the study's objectives, assured of confidentiality, and reminded that their participation was entirely voluntary.

In parallel with the interviews, the researchers conducted a comprehensive literature review and market analysis of existing systems such as high-beam assist and adaptive beams to identify core needs and technological gaps. The overarching purpose of the data collection was to obtain qualitative insights that would guide the development of the SmartBeam system, with a particular focus on understanding real-world night-time driving challenges and technical limitations in current vehicle systems. Specifically, the team sought to examine how the glare of oncoming headlights affects driver visibility, comfort, and safety; identify environmental conditions such as rain and fog that worsen the glare and reduce visibility; determine user expectations regarding system behavior, including response time and control; investigate the feasibility of integrating the system into different vehicle types; and assess the willingness of users to adopt and pay for the system. This information was essential for defining functional requirements such as automatic detection and dimming, non-functional requirements including affordability, durability, and usability, as well as entrepreneurial requirements.

3.3.1.1 Data collection methods employed

The primary method employed for data collection was interviews, which allowed the researchers to gather detailed, experience-based responses while maintaining consistency through a predefined interview guide. This method was particularly appropriate as it captured the real-life experiences of drivers dealing with glare and fatigue, enabled mechanics to explain technical challenges in detail, and provided the flexibility to probe deeper into specific responses. The interview guide was carefully designed and divided into two main sections: the driver section, focusing on night-time hazards, environmental conditions, and system expectations; and the mechanic section, focusing on hardware integration, sensor placement, and system durability.

3.3.1.2 Tool Refinement

Before actual data collection, the questions in the interview guide were reviewed to remove ambiguity, technical terms were simplified for better understanding, and some questions were adjusted following initial pilot interviews, ensuring that participants clearly understood the questions and provided meaningful responses.

3.3.1.3 Data collection procedures

The data collection process was carried out systematically over a period of two weeks, the process followed:

1. Identification of participants (drivers and mechanics)
2. Scheduling of interview sessions at convenient times
3. Conducting face-to-face interviews
4. Recording responses using structured notes and digital formats
5. Reviewing and organizing collected data for analysis

3.3.1.4 Ethical Considerations

Participants were informed about the purpose of the study, participation was voluntary, confidentiality of responses was maintained and no personal identifiers were included in the report

3.3.1.5 Data Recording Techniques

The researchers recorded the interviews using written notes and later categorized the responses into themes.

3.3.1.6 Sample population and size

The study targeted two main groups: drivers, who are the primary users affected by glare, and mechanics, who understand vehicle systems and installation challenges. These groups were selected because they provide complementary perspectives—drivers offering user experience insights and mechanics providing technical feasibility insights

3.3.1.7 Sampling Method

The researchers used convenience sampling, selecting participants based on availability and willingness to participate, a method deemed appropriate given the limited time and resources.

3.3.1.8 Sample Size

A total of 20 drivers and 5 mechanics participated in the interview study.

3.3.1.9 Justification

Although the sample size is small, it is sufficient for exploratory research, identifying key patterns and themes and guiding initial system design decisions

3.3.1.10 Method used for analysing data

For data analysis, the researchers applied thematic analysis to the interview responses to identify recurring ideas and patterns. The process involved grouping responses into categories such as glare impact, fatigue, and system expectations; combining similar responses into themes; and interpreting these themes to extract meaning. Thematic analysis was chosen because it is suitable for open-ended interview data and helps uncover hidden insights.

3.3.1.11 Tools Used

The tools used by the researchers to prepare for data collection included Microsoft Word for documenting interview findings and mobile phones for voice recording.

Other activities carried out during this phase included defining high-level functional requirements, such as oncoming light detection and dynamic intensity adjustment, as well as non-functional requirements like reliability in night, rain, and fog conditions, and modularity for compatibility. The researchers also selected initial hardware components, including an Arduino Board for rapid microcontroller prototyping, an ambient light sensor, a relay for headlight control, and a white light lamp to simulate headlight beams.

3.3.2 Phase 2: Build Initial Prototype

During this phase, the researchers developed a minimal working version of the system focused on core light detection and basic control. The hardware design involved assembling a breadboard circuit incorporating an Arduino Uno, ambient light sensors, and a relay module connected to simulated headlights. Software development comprised writing C++ code within the Arduino IDE for sensor data acquisition, implementing decision logic such that if oncoming light intensity exceeded a threshold, the system would reduce to low beam; otherwise, it would maintain high beam. Actuator control was achieved through relay switching for intensity levels.

3.3.3 Phase 3: Final Prototype Validation & Testing

The researchers conducted unit testing on individual components and performed integration testing, simulating oncoming headlights with controlled lights, the response time and accuracy where also measured.

3.3.3.1 Tools and Technologies

The tools and technologies the researchers used during the development of the final prototype included both hardware and software, the hardware tools included Arduino Uno, ambient light sensor, relay, PCB board, white light lamp for simulation, breadboards and a multimeter for testing. The software tool used was the Arduino IDE for writing the C++ code used to run the system

This prototyping approach ensured progressive risk reduction, early issue detection, and alignment with project objectives while producing a demonstrable, evaluated prototype within the academic constraints

Chapter 4

Requirements Analysis

4.1 Introduction

This chapter presents a comprehensive account of the requirements analysis conducted for the SmartBeam System, an integrated control system designed to automatically adjust vehicle headlight intensity based on environmental and traffic conditions.

The project addresses a critical road safety issue in Uganda that causes temporary blindness and reduced visibility caused by high-beam headlights during night driving.

The goal of this phase was to gather reliable data from both drivers and mechanics to ensure that the design of the system is user-centered, technically feasible and suitable for local conditions.

4.2 Data Analysis and Findings

The data collected from interviews with 20 drivers and 5 mechanics was analysed using thematic analysis. This section presents the findings organised according to the key themes that emerged from the responses of each participant group.

4.2.1 Analysis of Driver Responses

4.2.1.1 Theme 1: Prevalence of Night Driving

The majority of respondents reported that they frequently drive at night, mainly due to reduced traffic congestion and cooler environmental conditions for long journeys. This indicates that night driving is common and not occasional. Since night driving is frequent, the SmartBeam system must be reliable for continuous use rather than occasional scenarios. The system should operate efficiently over long durations without failure.

4.2.1.2 Theme 2: Severity of Glare

The study also indicated that most drivers rated the impact of glare between 7 and 10 on a scale of 1 to 10, with an average of approximately 8, indicating a high level of discomfort and danger. This high rating confirms that glare is not a minor inconvenience but a serious safety issue. It affects visibility and reaction time, increasing the likelihood of accidents. The researchers then concluded that automatic headlight dimming is not optional but a necessary feature.

4.2.1.3 Theme 3: Glare-Induced Fatigue

Drivers consistently reported that glare contributes significantly to fatigue and drowsiness, especially during long trips. Glare forces drivers to strain their eyes, leading to quicker exhaustion. This reduces concentration and increases the risk of human error. Therefore, the SmartBeam system should reduce glare instantly, improve visual comfort, and enhance driver alertness.

4.2.1.4 Theme 4: Real-World Safety Incidents

Several participants reported experiencing near accidents caused by bright headlights from oncoming vehicles. This provides real-world evidence that glare is directly linked to unsafe driving conditions. This provided strong justification that SmartBeam is a safety-critical system, not just a convenience feature.

4.2.1.5 Theme 5: User Expectations for System Behaviour

Drivers expressed that they would prefer:

i. Immediate response when detecting another vehicle
ii. Gradual dimming instead of abrupt switching
iii. Manual override option in case of system failure

The researchers interpreted these user preferences to mean that users want a system that is fast, smooth, and controllable.

4.2.1.6 Theme 6: Adoption and Cost Considerations

Most drivers indicated willingness to adopt the system only if it is affordable, meaning that cost is a major barrier in developing countries like Uganda. This made the researchers conclude that the system must be low-cost.

4.2.2 Analysis of Mechanic Responses

4.2.2.1 Theme 7: Vehicle Wiring Complexity

Mechanics highlighted differences between wiring older vehicles and modern vehicles, with older vehicles having simpler wiring systems compared to modern vehicles. This implies that the SmartBeam system must be designed with flexibility to accommodate both wiring configurations.

4.2.2.2 Theme 8: Technical Risks

Mechanics identified risks such as voltage fluctuations, incorrect wiring, and component damage. These risks necessitate the inclusion of protective features such as voltage regulators and fault-tolerant designs to ensure system longevity and safety.

4.2.2.3 Theme 9: Sensor Placement

Mechanics emphasized that sensor placement is critical for accurate detection, as poor sensor placement leads to incorrect readings and system failure. This highlights the need for clear installation guidelines and possibly adjustable mounting options.

4.2.2.4 Theme 10: Installation and Maintenance Preferences

Mechanics prefer systems that are easy to install, easy to repair, and modular in design. This preference directly informs the system's hardware architecture, encouraging a plug-and-play approach with replaceable components.

4.2.3 Summary of Key Themes

The overall interpretation of findings clearly shows that:

i. Glare is a major safety issue affecting visibility, comfort, and alertness

ii. There is strong demand for a solution that addresses these challenges

iii. Technical implementation is feasible, provided that design considerations account for vehicle variability, environmental conditions, and common technical risks

iv. The system must be cost-effective and easy to use to encourage widespread adoption in the Ugandan context

4.2.4 Relation of data to system requirements

The Table 4.1 clearly demonstrates how the findings from the data collection directly informed the system requirements of the SmartBeam system. Each requirement is based on real user experiences and technical insights. For example, the high severity of glare reported by drivers led to the inclusion of automatic light detection and intelligent dimming, while environmental challenges such as rain and fog influenced the need for high reliability and sensor fusion. Similarly, cost concerns and installation challenges informed non-functional requirements such as affordability and ease of installation.

This ensures that the final system design is both user-centred and technically feasible.

Table 4.1: Relationship of data to system requirements

Data Finding (From Analysis)	**System Requirement**	**Type**	**How It Informs System Design**
Drivers rated glare very high (avg = 8/10)	Automatic detection of oncoming light	Functional	Since glare is severe, the system must continuously detect incoming headlights using sensors to respond in real time and reduce brightness automatically.
Drivers experience temporary blindness and reduced visibility	Intelligent headlight dimming	Functional	The system should dynamically adjust light intensity (not just ON/OFF) to maintain visibility while preventing glare for both drivers.
Drivers reported near-accidents due to glare	Fast response time	Non-functional	The system must react instantly (within milliseconds) when detecting another vehicle to prevent dangerous delays that could lead to accidents
Majority reported fatigue due to glare	Intelligent dimming + reliability	Functional / Non-functional	The system must provide smooth and consistent dimming to reduce eye strain and ensure continuous operation without failure during long drives
Drivers prefer manual control option	Manual override control	Functional	A manual switch/button must be included to allow drivers to take control in case the automatic system fails or behaves unexpectedly.

Continued on next page

Table 4.1 – continued from previous page

Data Finding (From Analysis)	**System Requirement**	**Type**	**How It Informs System Design**
Glare worsens in rain and fog	High reliability	Non-functional	The system must work accurately under different environmental conditions using multiple sensors (sensor fusion) to avoid incorrect readings.
Mechanics reported voltage fluctuations	Durability	Non-functional	The system must include voltage regulators and protective circuits to withstand unstable power conditions in vehicles
Mechanics highlighted heat, dust, vibration issues	Durability	Non-functional	Components must be enclosed in protective casing (waterproof and heatresistant) to ensure longterm operation in harsh environment

4.3 Requirements Specification

To establish a clear link between user expectations and system execution, every requirement is indexed with a unique identifier.

4.3.1 Functional Requirements

Functional requirements specify the technical actions, computations, and transformations the embedded system must execute when responding to external environment stimuli as shown in Table 4.2.

Table 4.2: Functional Requirements

Requirement ID	Description	Input / Trigger	System Action
FR-01	Oncoming Light Detection	Light sensors sense illuminance thresholds	The system processes analog lux levels continuously
FR-02	Automatic Dimming Control	Lux input exceeds the pre-set high-beam threshold	The system switches the headlight from high beam to low beam
FR-03	Adaptive Beam Restoration	Lux input drops below the safe ambient threshold	The system restores headlights back to high beam automatically

4.3.2 Non-Functional Requirements (NFR)

Non-functional requirements specify the performance criteria, constraints, and quality attributes of the system.

NFR-1: Performance (Latency) – The system must adjust headlight intensity within 1 second of detecting oncoming glare.

NFR-2: Electrical Safety & Reliability – The circuit must protect microcontroller pins from overcurrent using inline resistors and optocouplers.

NFR-3: Affordability – The total production cost of the retrofit package must remain low-cost for local drivers.

NFR-4: Maintainability – The hardware design must be modular to ensure fast component replacement by mechanics.

Chapter 5

System Design

5.1 Introduction

This chapter presents the detailed System Analysis and Design for the SmartBeam system. It translates the empirical qualitative findings from field interviews with drivers and mechanics into structured technical specifications.

The documentation proceeds from system requirements gathering (defining what the system does) to logical design (abstract blueprints) and physical design (concrete hardware blueprints, circuits, and programmatic architectures).

5.2 Logical Design

The logical design outlines the abstract data flow and operational models without dependency on specific hardware.

5.2.1 Data Flow Diagram (DFD) - Level 0 (Context Diagram)

The Level 0 Data Flow Diagram (Context Diagram), illustrated in Figure 5.1, defines the conceptual boundaries of the SmartBeam system. It tracks how physical light metrics are transformed into localized electrical routing and programmatically managed states across multiple structural processes and data repositories.

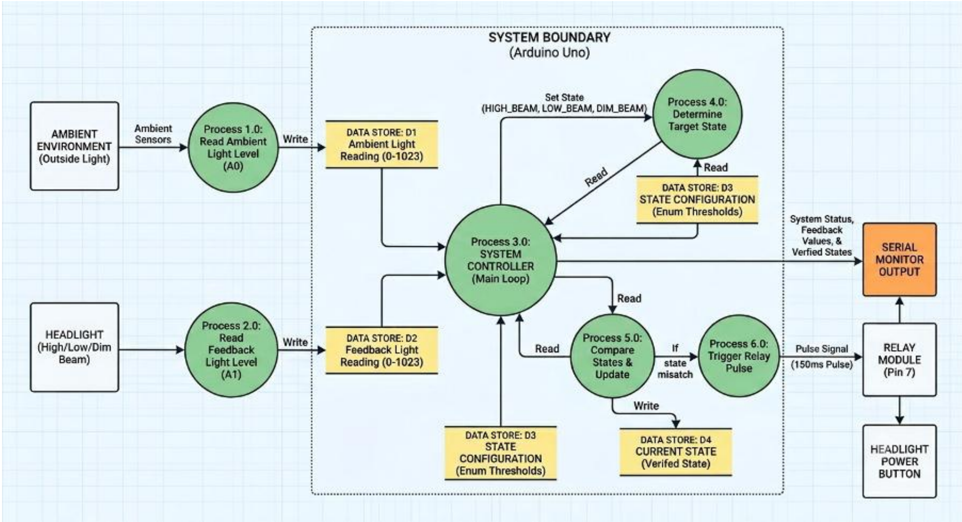

Figure 5.1: Data flow diagram

The data flow within the system boundary operates through five primary sequential stages:

i. **Sensing and Data Input Transformations:** The system continuously samples two separate external hardware inputs. Process 1.0 (Read Ambient Light Level) monitors the outside light using front-facing ambient sensors to look for oncoming high-beams or ambient infrastructure light. Concurrently, Process 2.0 (Read Feedback Light Level) reads the localized luminosity directly from an isolated subsection of the host headlight lens. Both processes convert these physical variations into 10-bit analog numbers ranging from 0 to 1023.

ii. **Data Serialization and Storage:** Once converted, these values are committed to temporary memory registers. Process 1.0 writes its calculated metrics into Data Store D1 (Ambient Light Reading), while Process 2.0 writes its metrics into Data Store D2 (Feedback Light Reading). These repositories act as buffering points that feed fresh, empirical lighting benchmarks to the core application loop.

iii. **Central Processing Loop:** Process 3.0 (System Controller / Main Loop) forms the central engine of the system architecture. It constantly pulls the filtered data from Data Stores D1 and D2, acting as the main interface between incoming information and the execution steps. Simultaneously, Process 3.0 outputs raw tracking telemetry to the Serial Monitor Output for real-time diagnostic auditing and verification.

iv. **State Determination and Verification Logic:** The System Controller hands off execution metrics to Process 4.0 (Determine Target State) and Process 5.0

(Compare States & Update). Process 4.0 cross-references ambient values with programmed parameters inside Data Store D3 (State Configuration) to flag the target light mode (HIGH_BEAM, LOW_BEAM, or DIM_BEAM). Process 5.0 then compares this calculated target against the active hardware profile sitting inside Data Store D4 (Current State). If the current hardware deployment matches the calculated ideal setting, the loop finishes without executing further tasks.

v. **Isolated Actuation and Loop Closure:** If Process 5.0 encounters an error flag via a State Mismatch, it passes an execution order to Process 6.0 (Trigger Relay Pulse). This process converts the logic error into a physical fix by sending a timed, Active-LOW Pulse Signal (150ms Pulse) out from Digital Pin 7. This pulse hits the physical Relay Module, bridging the headlight's sequential button circuit to execute a hardware transition. The physical results are immediately registered back into Process 2.0 via the internal optical capture housing, successfully locking the closed-loop architecture.

5.2.2 Use Case Modelling

The system features one primary automated process interacting with two key operational triggers.

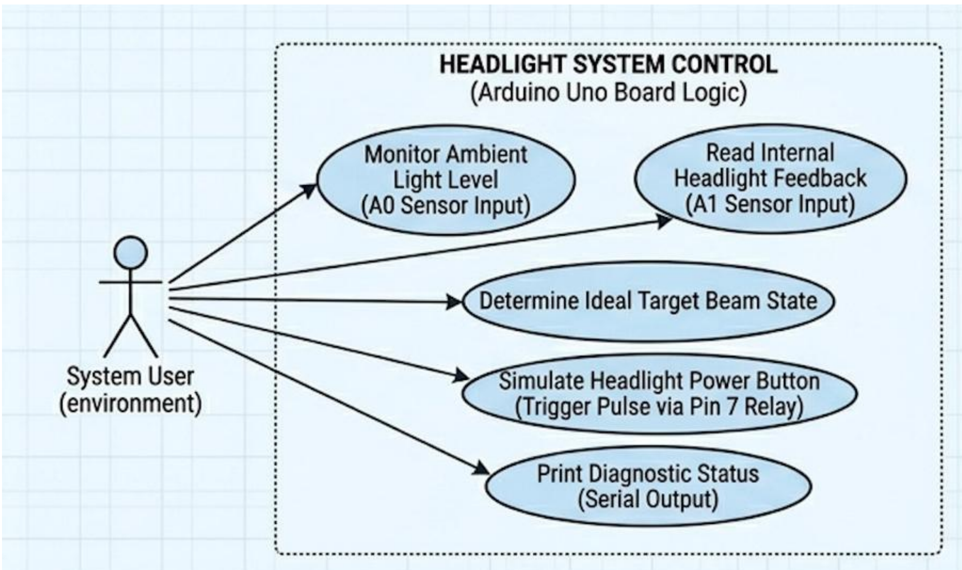

Figure 5.2: Use case of user environment

In the Figure 5.2, Core System Functions, the system's operational capabilities are defined within the boundary of the Arduino Uno microcontroller. The primary actor is designated as the System User (Environment), which represents the dynamic real-world driving conditions surrounding the vehicle. This actor interacts with the system through two distinct data input vectors: the continuous sampling of road brightness via the Monitor Ambient Light Level (A0 Sensor Input) use case, and the physical verification of headlight luminosity via the Read Internal Headlight Feedback (A1 Sensor Input) use case. Based on these concurrent inputs, the Arduino executes the Determine Ideal Target Beam State use case to compute the optimal light configuration. To execute physical changes, the system interacts with the headlight hardware through the Simulate Headlight Power Button use case, which outputs a precise timing pulse via a Digital Pin 7 relay. Concurrently, system visibility is maintained through the Print Diagnostic Status use case, which outputs raw values and verified states to the serial terminal for real-time monitoring.

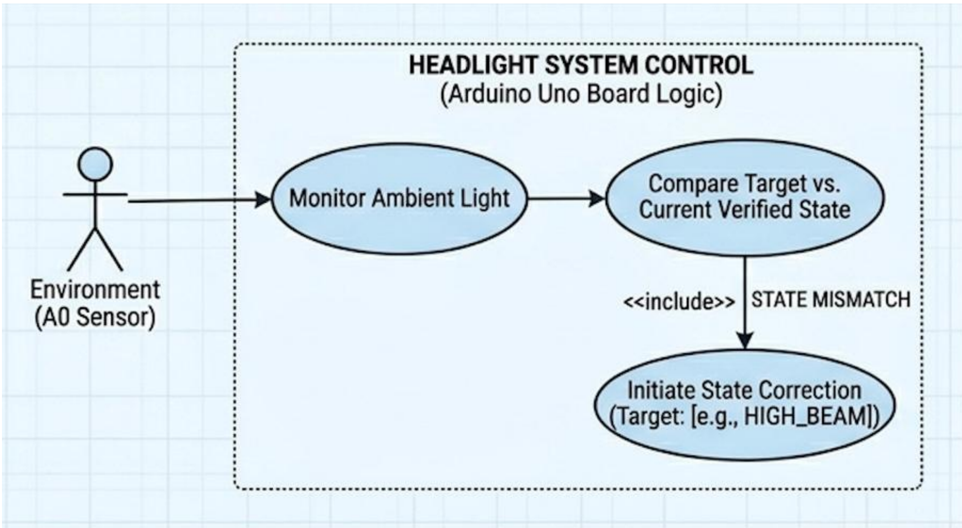

Figure 5.3: Use case of light sensor at analog pin 0

In Figure 5.3, Adaptive Lighting Logic shifts focus toward a specific behavioral workflow that dictates system corrections. Triggered directly by environmental changes picked up by the A0 sensor, the microcontroller executes the Monitor Ambient Light use case, passing the data directly to the Compare Target vs. Current Verified State logic block. This step evaluates the calculated environmental demand against the actual state reported by the feedback sensor. If the states match, the operation concludes; however, if a discrepancy is found, it triggers a mandatory <<include>> relationship labeled STATE MISMATCH. This architectural dependency forces the system to call upon the Initiate State Correction use case. This function actively locks the system into an automated correction cycle, rapidly pulsing the relay module until the physical headlight matches the required target state, thereby closing the control loop securely.

5.3 Physical Design

The physical design details how the logical structures are mapped to concrete hardware components and electrical architectures.

5.3.1 Hardware Architecture & Technology Stack

The hardware implementation is built on robust, accessible microcontroller technologies, these include:

i. Control Unit: Arduino Uno board running optimized C++ compilation code

ii. Sensing Array: High-sensitivity Light Dependent Resistors (LDRs) with multipoint sensor fusion layout.

iii. Actuation Module: 5V Relay module switches headlight beam state.

iv. Development Environment: Arduino IDE utilizing open-source microcontroller packaging utilities.

5.3.2 System Circuit Diagram

The hardware implementation translates environmental and internal light metrics into localized electrical routing using three distinct, interconnected functional blocks. As illustrated in the system circuit diagram in Figure 5.4, these subdivisions isolate delicate low-voltage processing logic from high-current vehicle electrical loads while establishing a stable signal path.

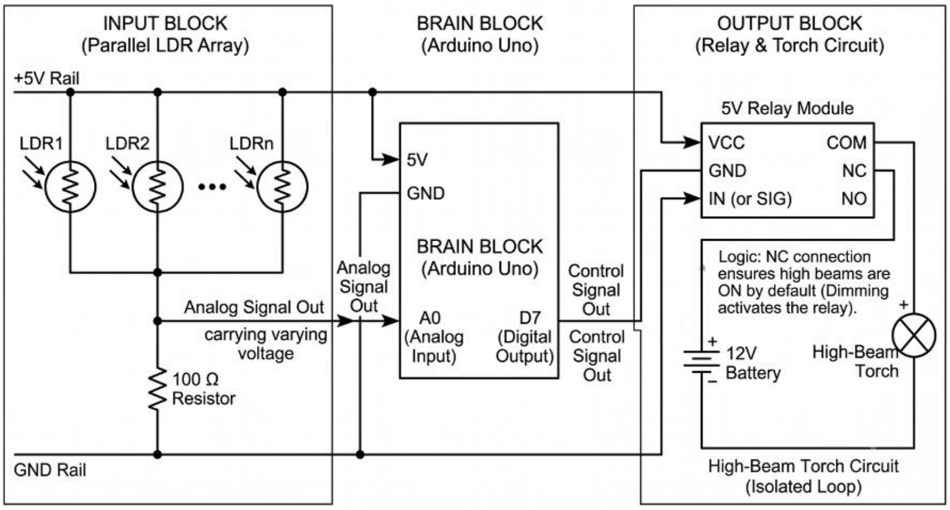

Figure 5.4: Circuit diagram showing electricity flow between the components

Chapter 6

System Implementation and Testing

6.1 Introduction

This chapter details the physical implementation, prototyping assembly, component interaction mechanics, and empirical testing phases of the SmartBeam system. It documents the critical engineering transition from abstract schematic designs to physical hardware validation. The following sections outline the structured hardware wiring layout, the closed-loop component interaction model, the software calibration matrices, and the testing protocols used to verify that the system satisfies all operational, functional, and safety requirements.

6.2 Hardware Assembly and Prototyping

The hardware implementation was executed by constructing a functional bench-top prototype designed to simulate a vehicle's electrical environment under controlled conditions.

6.2.1 Component Wiring Matrix

To prevent accidental short-circuits, manage voltage step-downs, and ensure clean signal routing, physical connections were mapped according to a strict hardware wiring configuration. Table 6.1 outlines the structural linkages between the processing, sensing, and actuation sub-systems.

Table 6.1: Component wiring

Source Component	**Pin Out**	**Target Component**	**Pin In**
Buck Regulator (5V)	Vout AWG Solid Jumper	Arduino	5V Pin22
LDR Sensor Array	Signal Out	Arduino	Pin A0
Arduino	Pin 7	Relay Module	Signal/In
Relay Output (COM)	Common	Vehicle Battery (12V)	positive (+)
Relay Output (NC)	Normally Closed	Headlamp Module	High-Beam Pin
Relay Output (NO)	Normally Open	Headlamp Module	Low-Beam Pin

6.2.2 Component Interaction Mechanics

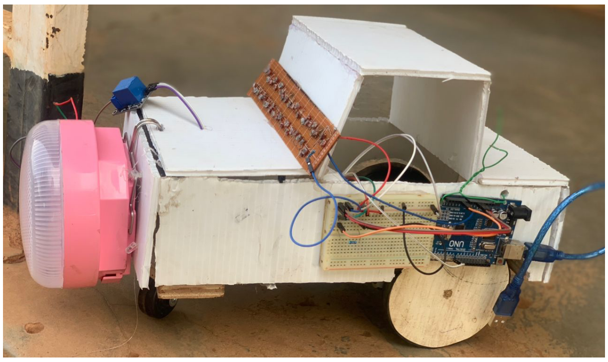

Figure 6.1: Showing the system prototype and the component interactions

The system functions as a unified, closed-loop automation platform where data inputs directly modulate mechanical outputs as shown in Figure 6.1, which are then instantly re-verified by optical inputs. The core interaction flow operates as follows:

1. **Dual-Vector Sensing:** The system splits its sensory input into two vectors. The Ambient LDR (Pin A0) monitors the external environment for oncoming vehicle beams or streetlights. Simultaneously, the Feedback LDR (Pin A1) is physically isolated within a light-tight capture housing pointed directly at a sub-section of the headlamp lens to sample its real-time luminous intensity.

2. **Control Processing:** The Arduino Uno compares the ambient value against its light thresholds. If a change is required, it checks the internal feedback loop to

evaluate the current state. If a state mismatch exists, Pin 7 sends an execution signal to the relay module.

3. **Isolated Actuation:** The relay module functions using Active-LOW configuration logic for rapid engagement. When Pin 7 triggers a brief 150ms pulse, the relay's internal electromagnet forces the internal switch contacts closed. Because the Common (COM) and Normally Open (NO) terminals are soldered directly across the contacts of the headlight's sequential mode button, this pulse mimics a rapid physical finger press.

4. **Loop Closure:** Immediately following the pulse, the headlight changes its hardware mode (e.g., transitioning from High Beam to Low Beam). The Feedback LDR captures this immediate alteration in internal brightness and passes the updated analog value back to the Arduino, confirming the state transition was physically executed before the system resumes road monitoring.

6.3 Software Implementation and Calibration

The core software code of the system was developed using C++ within the Arduino IDE as shown in Figure 6.2 below, the software layout executes a non-blocking, continuous polling loop. This loop actively mitigates ambient noise while managing state-correction cycles through a strict, data-driven framework.

To ensure distinct state isolation, a pull-down resistor optimization study was performed. The final system incorporates a 2.2 kΩ resistor for the feedback LDR voltage divider, widening the voltage separation between the headlight's compressed operational beam profiles. This allows for stable, deterministic processing thresholds as outlined in Table 6.2.

Table 6.2: Showing the LDR sensor's interaction with the relay

Empirical Light Level	**Raw ADC Values Range**	**Programmatic Threshold Boundary**	**Defined Software State (TorchState)**
Dark Baseline	0 to 20	feedbackDimMax = 220	OFF
Dim Luminous Profile	21 to 499	feedbackLowMax = 500	DIM_BEAM
Low Beam Profile	500 to 619	feedbackHighMax = 620	LOW_BEAM
High Beam Profile	620 to 1023	Upper Ceiling	HIGH_BEAM

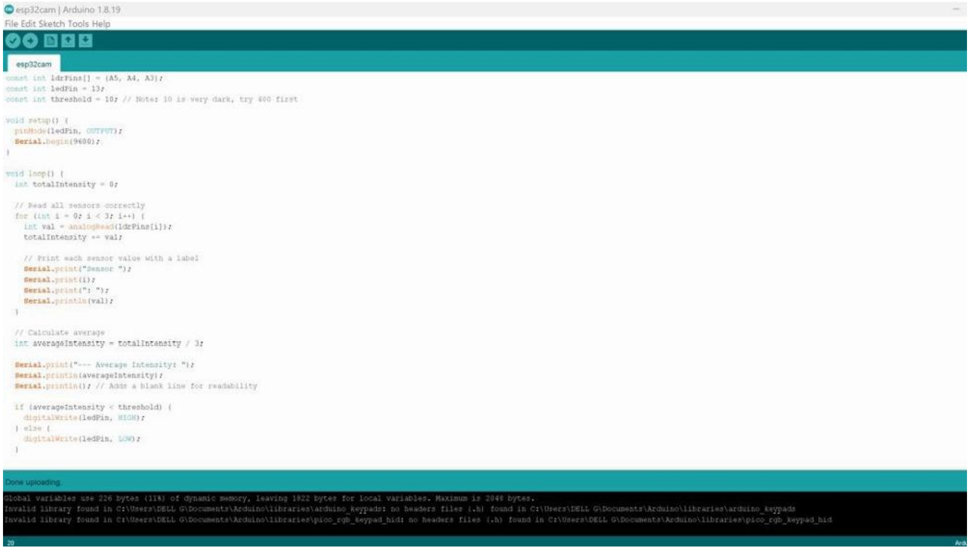

Figure 6.2: Arduino code for the system

z

6.4 System Testing Protocols and Validation

Validation was conducted through iterative laboratory simulations to ensure the system safely handles transition logic, avoids infinite cycling loops, and adapts correctly to sudden changes in environmental illumination.

6.4.1 Diagnostic Test Phase 1: Open-Loop Blind Testing

Initial testing checked how the system tracked states using software code increments without checking sensor inputs.

i. **Protocol:** The system was booted up, and the ambient LDR (A0) was covered to simulate dark driving conditions, forcing a target state of HIGH_BEAM. The sensor was then blasted with light to trigger a target state of DIM_BEAM.

ii. **Observations:** The serial monitor registered the state requests accurately. However, if the headlight missed a relay pulse due to minor connection latency, or if the headlight was manually turned on by hand, the code continued tracking an incorrect state variable. The software became "blind" to the hardware reality, leaving the system completely out of phase with the actual headlight beam output.

6.4.2 Diagnostic Test Phase 2: Closed-Loop Dynamic Tracking Validation

The closed-loop architecture featuring the rapid state-correction while loop and the 2.2 kΩ feedback sensor was subjected to rigorous stress-testing.

i. **Protocol:** The headlight was intentionally turned off before booting up the Arduino to create a severe state misalignment. The ambient sensor (A0) was placed in total darkness, enforcing a HIGH_BEAM target.

ii. **Observations:** As detailed in the log excerpt above, the system immediately intercepted the mismatch upon boot-up. Instead of waiting for a slow loop cycle, the code entered its rapid while loop, pulsing the Active-LOW relay module. It stepped through OFF (ADC: 85), DIM_BEAM (ADC: 446), and LOW_BEAM (ADC: 570), reading the real-time optical feedback every 250 ms Upon the fourth execution pulse, the headlight hit its maximum intensity (ADC: 676). The Arduino verified this as HIGH_BEAM, successfully exited the correction loop, and safely paused further relay actions.

6.4.3 Test Summary and Environmental Resilience

Final verification verified that isolating the feedback LDR inside a sealed capture tube eliminated interference from external ambient lighting changes. Even when strong overhead laboratory lights were turned on, the feedback sensor readings remained stable within their calibrated ranges. The 250ms software delay gave the internal headlight electronics plenty of time to fully power up and stabilize their light output before the Arduino took an analog reading, preventing incorrect readings caused by voltage fluctuations during switching. This confirms that the system operates with the high reliability needed for automotive safety applications.

It is important to acknowledge that testing the SmartBeam system on an actual passenger vehicle under real-world driving conditions presented significant practical challenges. Access to a suitable test vehicle, the logistical complexity of integrating the prototype into a car's electrical system, and safety concerns associated with on-road testing during night-time hours made real-vehicle testing infeasible within the project's academic timeline and resource constraints. Consequently, the research team opted for a controlled laboratory environment that faithfully simulated the essential lighting conditions and electrical characteristics of a vehicle's headlight system. While this approach provided a rigorous and repeatable framework for validating the system's core functionality, response time, and accuracy, it does not fully replicate the complex environmental factors such as road vibrations, varying weather conditions, and the dynamic nature of oncoming traffic that the system would encounter in actual deployment. Future work should therefore prioritize field testing on real vehicles to further validate the system's long-term durability and real-world effectiveness.

Chapter 7

Discussions, Conclusions and Recommendations

7.1 Introduction

This chapter presents a discussion of the findings obtained throughout the development and testing of the SmartBeam system, reflects on the extent to which each specific objective was achieved, and offers recommendations for improvement and future development. It concludes with a summary statement affirming the overall contribution of the project.

7.2 Discussion of Findings

7.2.1 Objective One: To Investigate the Current Effectiveness of Existing Technologies

The investigation into existing headlight technologies was conducted through a comprehensive literature review and semi-structured interviews with 20 drivers and 5 mechanics. Findings from the literature revealed that while advanced systems such as matrix headlights and adaptive driving beams offer superior visibility and precision, they remain expensive and limited to premium vehicle segments (Musthafa et al., 2017). High-beam assist systems, which use forward-facing cameras to detect oncoming traffic, operate on a binary all-or-nothing approach that temporarily reduces the host driver's long-range visibility (Balaji, 2020). Adaptive headlights improve visibility on curved roads but introduce mechanical complexity and higher maintenance costs (Musthafa et al., 2017).

The interview data confirmed that glare is a severe problem in Uganda, with drivers rating its impact at an average of 8 out of 10. Several participants reported experiencing near-accidents caused by bright headlights. Mechanics highlighted that the wiring complexity of older vehicles, which constitute the majority of the Ugandan fleet, makes retrofitting advanced systems impractical and cost-prohibitive. The investigation concluded that while effective technologies exist, there is a significant gap in affordable, practical solutions suitable for the majority of vehicles on Ugandan roads.

7.2.2 Objective Two: To Design an Embedded Program that Uses Advanced Sensor Technology

The design of the SmartBeam system was guided by findings from the requirements gathering phase. The logical design was captured through Data Flow Diagrams and Use Case Models, defining how physical light metrics are transformed into electrical routing and programmatically managed states. The physical design mapped logical structures to concrete hardware components, utilizing an Arduino Uno microcontroller for its affordability, Light Dependent Resistors (LDRs) for light sensing, and a 5V relay module for actuator control. Two LDRs were incorporated: one to monitor ambient light and a second to provide closed-loop feedback by reading the actual state of the headlight.

The software design followed a continuous polling loop architecture written in C++ within the Arduino IDE. The decision logic was structured such that if oncoming light intensity exceeded a pre-set threshold, the system would reduce to low beam; otherwise, it would maintain high beam. User expectations identified during interviews, such as gradual dimming instead of abrupt switching and manual override capability, were incorporated into the design. The closed-loop architecture ensures the system validates every physical action, addressing the tracking desynchronization identified in open-loop systems.

7.2.3 Objective Three: To Implement the Designed Embedded Program for Automatic Headlight Adjustment

The implementation phase translated the design specifications into a fully functional prototype. The hardware was assembled on a breadboard incorporating an Arduino Uno, LDR sensors, and a relay module connected to simulated headlights. A 2.2 k pull-down resistor was selected for the feedback LDR voltage divider to widen the voltage separation between the headlight's operational beam profiles. The software was written in C++ within the Arduino IDE and uploaded to the Arduino board.

The software implementation centered on two key components: a feedback tracking mechanism and a state-correction loop. The feedback system maps internal light intensity to four states (OFF, DIM, LOW, HIGH). The state-correction loop pulses the relay when a mismatch occurs until the headlight matches the target state. A 250ms delay prevents voltage fluctuation errors. The implementation produced a working prototype for automatic headlight adjustment.

7.2.4 Objective Four: To Test the Effectiveness of the Embedded Program

Testing was conducted through rigorous laboratory simulations to verify system functionality and reliability. Initial open-loop testing revealed a critical limitation: if the headlight missed a relay pulse or was manually turned on, the software continued tracking an incorrect state variable, leaving the system completely out of phase with the actual headlight beam output. This confirmed the need for closed-loop tracking.

The closed-loop architecture was rigorously tested by intentionally creating a state misalignment. The system immediately detected the mismatch and entered a correction loop, cycling through beam states until reaching the target. The loop successfully exited upon verification. Final tests confirmed that isolating the feedback sensor eliminated external light interference, and the 250ms delay prevented incorrect readings during switching.

Response time measurements confirmed that the system reacts within milliseconds of detecting oncoming light. Accuracy testing demonstrated that the system correctly identifies the appropriate beam state in all tested scenarios under controlled conditions. The manual override function was verified to work as intended. The testing confirmed that SmartBeam successfully mitigates tracking desynchronization and operates with the high reliability needed for automotive safety applications.

7.2.5 Research Contributions

This project makes several meaningful contributions to the field of embedded systems and automotive safety in Uganda. The first contribution is the development of a context-specific automatic headlight adjustment system designed explicitly for vehicles in Uganda, built from data collected directly from local drivers and mechanics. The second contribution is the design and implementation of a closed-loop control architecture using affordable components, proving that advanced safety features can be implemented cost-effectively. The third contribution is the sensor calibration methodology developed to optimize LDR-based sensing, providing a replicable framework for similar applications. The fourth contribution is empirical evidence confirming that glare is a severe safety issue in Uganda, adding to the limited body of research on road safety challenges in the local context. Finally, the project contributes a fully documented prototype including circuit diagrams, software code, and hardware specifications that can serve as a practical reference for future embedded systems development.

7.2.6 Limitations of the Study

The project was subject to several limitations. The sample size of 20 drivers and 5 mechanics, while adequate for exploratory research, limits the statistical generalizability of the findings. The convenience sampling method introduced potential bias. The prototype was evaluated under controlled laboratory conditions rather than real-world driving environments, so aspects such as long-term durability and real-world reliability were not comprehensively tested. The study was confined to a short time scope, making it impossible to measure the system's long-term reliability or impact on accident reduction. The reliance on LDRs creates vulnerability to complex lighting environments, and a cost-benefit analysis was not conducted.

7.3 Conclusion

The SmartBeam project has successfully demonstrated that an innovative and highly affordable embedded system can effectively combat the critical hazard of night-time headlight glare on Ugandan roads. By utilizing an Arduino Uno, responsive light sensors, and a relay module, the fully working system delivers a reliable, low-latency solution that automates beam adjustments in real-time. This project proudly proves that sophisticated automotive safety features do not have to be exclusive to luxury cars. Instead, SmartBeam provides a practical, cost-effective retrofit for existing passenger vehicles, offering an impactful, life-saving deployment that drastically reduces night-time accidents and elevates road safety across Uganda.

7.4 Recommendations for future development

To further build upon the great success of this fully functional system and maximize its real-world impact, the following future enhancements are highly recommended:

i. Adopt Custom Printed Circuit Boards (PCBs): The current prototype relies on physical wires and breadboard connections, which, while effective for proof-of-concept, present challenges in terms of bulkiness, potential connection looseness, and aesthetic integration. Future iterations should transition to custom-designed Printed Circuit Boards (PCBs) that can be compactly integrated directly into the vehicle's headlight housing or electronic control unit. A PCB-based design would not only produce a cleaner, more professional appearance but also enhance reliability by eliminating loose wire connections, reduce electromagnetic interference, simplify installation for mechanics, and lower manufacturing costs through scalable production. This approach would make the SmartBeam system significantly more feasible for commercial adoption and aftermarket installation in Uganda's passenger vehicles.

ii. Upgrade to Intelligent Sensors: Transition from basic light sensors to a compact camera module to allow the system to intelligently differentiate between oncoming vehicle headlights and static ambient lighting like streetlights.

iii. Implement Seamless Dimming: Replace the mechanical relay module with solid-state components (such as a MOSFET) to unlock smooth, continuous brightness adjustments instead of basic high-and-low beam switching.

iv. Integrate Vehicle Dynamics: Incorporate an angle or steering sensor to allow the SmartBeam system to proactively adjust headlight direction and intensity when navigating sharp curves, hills, or winding roads.

v. Incorporate Weather Adaptation: Add environmental sensors to automatically adjust the system's light thresholds during heavy rain or thick fog, keeping drivers safe across all seasonal conditions.

vi. Conduct Long-Term Field Testing: Expand the testing of the active system on actual passenger vehicles (such as sedans and SUVs) across diverse highways in Uganda to validate its long-term durability against continuous road vibrations and real-world weather patterns.

References

[1] Abdulrahman, R., Almoshaogeh, M., Haider, H., Alharbi, F., & Jamal, A. (2025). Development and application of a risk analysis methodology for road traffic accidents. *Alexandria Engineering Journal*, *111*, 293–305.

[2] Akbar Gozali, A., et al. (2025). Smart train control and monitoring system with predictive maintenance and secure communications features. *Transportation Research Interdisciplinary Perspectives*, *31*, 101409.

[3] Balaji, R. D. (2020). A case study on automatic smart headlight system for accident avoidance. *International Journal of computer communication and informatics*, 2(2), 70-77.

[4] Joseph, S. S., et al. (2026). Scoping review on motorcycle crashes patterns, risk factors, and potential in setting policy priorities in the gulf cooperation council countries (GCC). *Injury*, *57*(3), 113017.

[5] Li, H., Wang, L., Yang, M., & Bie, Y. (2025). Collaborative effects of vehicle speed and illumination gradient at highway intersection exits on drivers' stress response capacity. *Accident Analysis & Prevention*, *209*, 107829.

[6] Moradloo, N., Mahdinia, I., & Khattak, A. J. (2025). Nighttime safety of pedestrians: The role of pedestrian automatic emergency braking systems. *Accident Analysis & Prevention*, *219*, 108110.

[7] Musthafa, R. A., Bala Krishnan, T., Seetha Raman, N., Shankar, M., & Swathi, R. (2017). Automatic headlight beam controller. *Int J Trend Res Dev (IJTRD).*

[8] Naskar, A. K., & Saha, P. (2025). Controlled experiments for road safety assessment – A conceptual framework. *Next Research*, *2*(2), 100277.

[9] Raychaudhuri, S., Challa, T., & V, J. (2025). Challenges in the design & development of efficient (High Power) lighting systems. *Optik*, *338*, 172495.

[10] Xie, X., Hu, Y., Feng, J., Xu, W., & Li, M. (2025). A reflective method for the optical axis calibration of high-power lasers. *Measurement*, *246*, 116714.

[11] Zhou, L., Feng, S., Kan, D., & Mao, W. (2025). Investigation on effect of pavement material on obstacle visibility and lighting energy consumption in road tunnel. *Tunnelling and Underground Space Technology*, *156*, 106249.

Appendix A

Hardware Components

Table A.1: Hardware components

Category	**Component**	**Purpose**
Controller	Arduino uno	Main Brain
Power	5V USB Power, 7-12V Battery Pack	System power
Input	Light sensor	Detecting light intensity
Actuator	Relay module	Adjust Headlight intensity
Lights	White lamps (x2) + Resistors	Simulate High/Low Beams
Prototyping	Breadboard, Jumper Wires	Build Circuit
Structure	Cardboard/Acrylic, Glue, Tape	Build Model Car
Software	Arduino IDE, Arduino Board Package	Write & Upload Code

Appendix B

Driver Interview Guide

Section A: Nighttime Hazards

1. How often do you drive at night?
2. On a scale of 1–10, how does glare affect you?
3. Does glare increase fatigue?
4. Have you experienced a near accident due to headlights?

Section B: Environmental & System Requirements

1. Does rain or fog make glare worse?
2. How fast should lights dim automatically?
3. Is manual override important?
4. Do you prefer gradual dimming or instant switching?

Section C: Adoption & Cost

1. Would you install this system if affordable?
2. How much would you be willing to pay?

Appendix C

Mechanic Interview Guide

Section A: Electrical Integration

1. What are common wiring systems in vehicles?
2. What risks exist with microcontrollers (e.g., Arduino Uno)?
3. What challenges exist with relays/H-bridges?
4. What voltage issues occur in motorcycles?

Section B: Sensor Placement

1. Where should sensors be mounted?
2. How can hardware be protected from heat/vibration?
3. What waterproofing methods are best?

Section C: Technical Design Feedback

1. Is sensor fusion better than a single sensor?
2. What makes the system easy to install and repair?

Appendix D

System Images

GULU UNIVERSITY

Faculty of Science

OFFICE OF THE DEAN FACULTY OF SCIENCE

9th March 2026

To: Research participants

Dear Sir/Madam,

RE: INTRODUCTION OF Kabagyo Desire Acquilina 23/U/1626/GCS/PS, Lanyero Fortunate 23/U/3718/GCS/PS AND OAkejo Ivan Gira 23/U/1605/GCS/PS -AN EMBEDDED PROGRAM FOR AUTOMATIC HEADLIGHT INTENSITY ADJUSTMENT

(SMARTBEAM)

I am pleased to introduce the individuals named above, who are students in the Department of Computer Science at Gulu University, currently pursuing a Bachelor of Science degree in Information and Communication Technology.

As part of their academic requirements, the candidate is undertaking research titled ***"AN EMBEDDED PROGRAM FOR AUTOMATIC HEADLIGHT INTENSITY ADJUSTMENT***

(SMARTBEAM)" To successfully carry out this study and gather accurate system requirements, the student intends to engage with key personnel from your organization through questionnaires, focus group discussion and questionaires. These interactions, including data collection and evaluation of the system prototype, are expected to take up to two months.

The purpose of this letter is to formally request your organization's participation in this research by allowing the candidate to access relevant information and interact with appropriate staff. Your insights and support will be crucial in ensuring the system is user-centered, contextually appropriate, and beneficial to the preservation of cultural heritage sites.

For further details regarding the research, you may contact the candidates directly via phone at0764555383.

Any assistance you render to the candidate will be greatly appreciated.

Sincerely,

Assc. Professor Malinga Geoffrey Maxwell
Dean Faculty of Science, Gulu University

Figure D.1: Introductory letter presented to the research participants

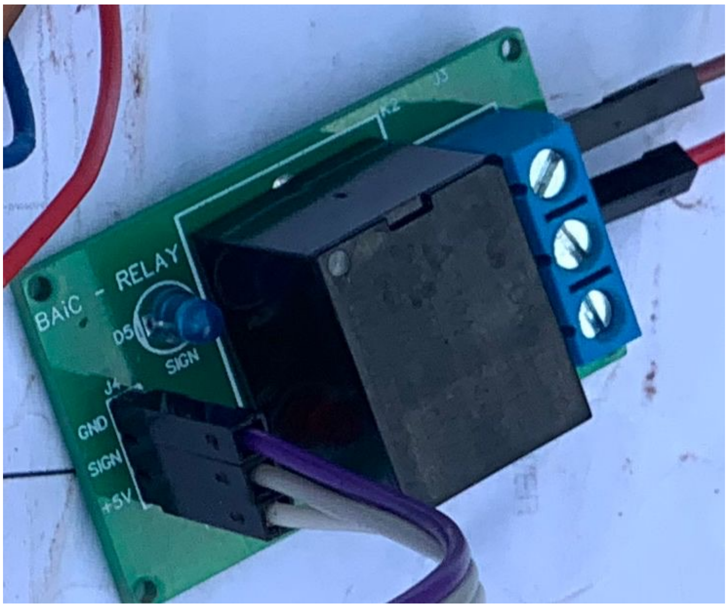

Figure D.2: Relay module for automatically switching between the light beams

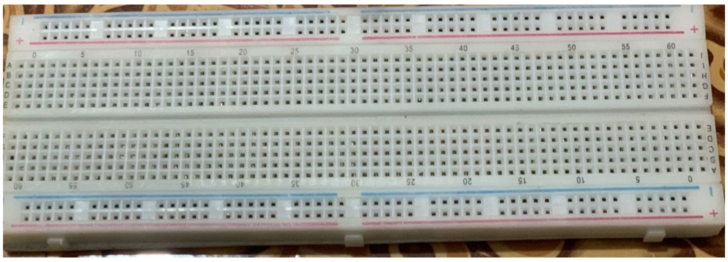

Figure D.3: Breadboard for making connections

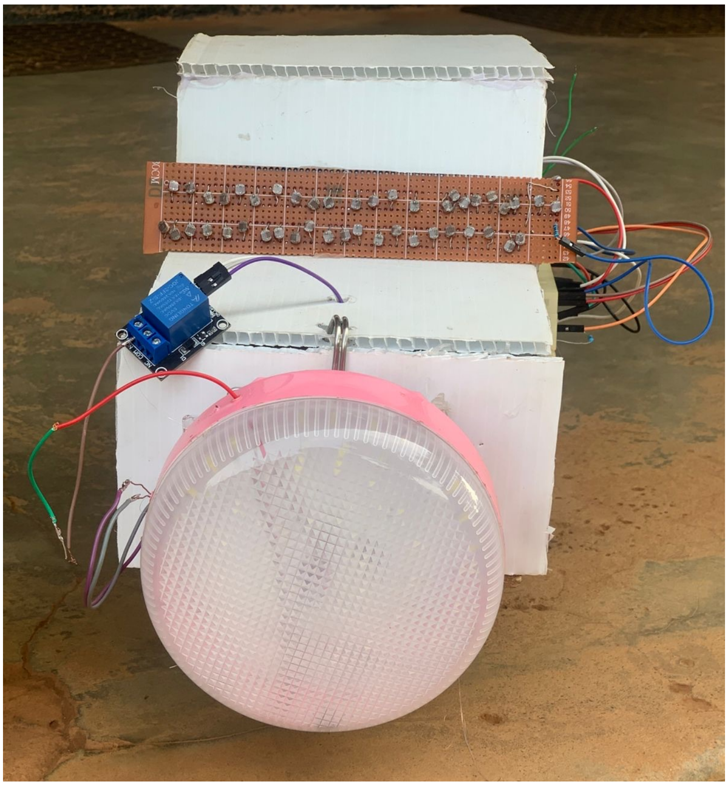

Figure D.4: Front view of the car prototype mounted with the SmartBeam system

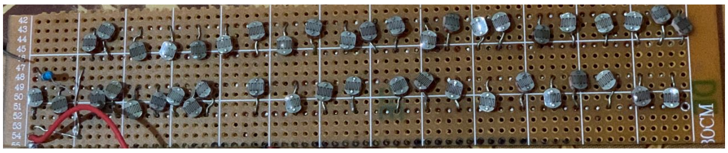

Figure D.5: Light sensor developed by connecting multiple LDR sensors in parallel to increase it's surface area

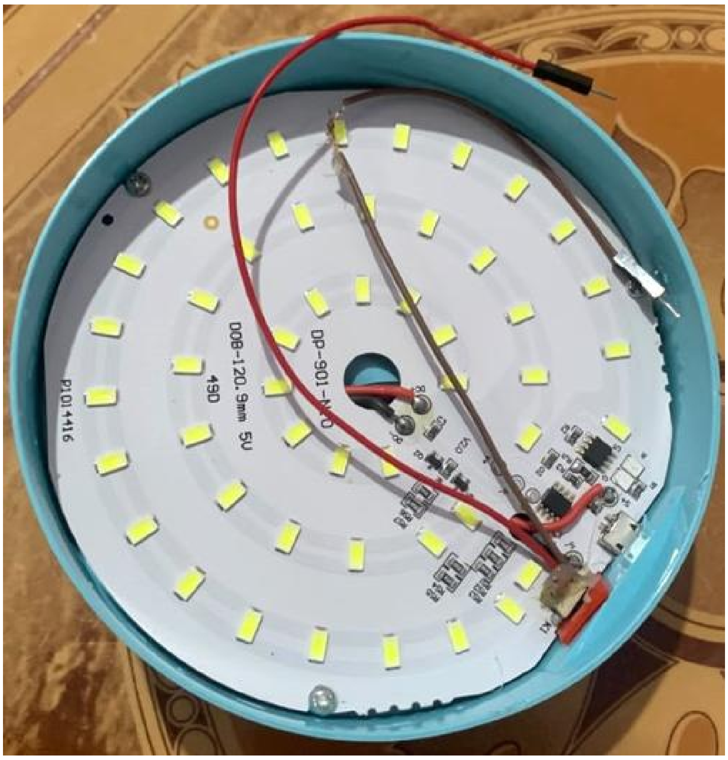

Figure D.6: White light lamp with connections to the switch that are connected to the relay module

Figure D.7: The researcher soldering the array of LDR sensors onto a PCB